RESUME

Name : Hoàng Bá Duy

Date of Birth: 1/9/2004

Home Address: 36 Dịch Vọng Hậu,Cầu Giấy, Hà Nội

Home Page : invencamp.com

Contact Number: 0357693159

Email : hbduy31@gmail.com

EDUCATION

PERIOD (YYYY/MM)	SCHOOL	MAJOR	GPA	BEST ACHIVEMENT
2022/09 – 2026/05	VNU University of Engineering and Technology	Mechatronics	3.15	GPA: 3.6 (Year 2, Semester 1)
2019/09 – 2022/05	An Thi High School			

S

EXPERIENCE

PERIOD (YYYY/MM)	COMPANY	TASK

SKILLS

CATEGORY	DESCRIPTION
LANGUAGE	English (B2 VNU-Tests), TOEIC IIG 730
PROGRAMING	C++ (★★★★☆)
	Python (★★★★☆)
HARDWARE	Arduino (★★★★☆)
	ESP32 (★★★☆☆)
	STM32 (★★★☆☆)
	ATmega16 (★★★★☆)

FRONT - END	HTML (★★☆☆☆) NodeJS (★★★☆☆)

CERTIFICATE & AWARDS

CATEGORY	DESCRIPTION
CERTIFICATE	B2 VNU-Tests, TOEIC IIG 730
AWARDS	Samsung Innovation Campus 2025, One Day Boshcher 2025,

PROJECT

PERIOD (YYYY/MM)	DESCRIPTION
2023/09 - 2023/12	**Automated car solving maze by DFS and BFS algorithm** - Using Arduino to controll 4 DC motor - C Program (Arduino IDE) - PID Control - ultrasonic sensor
2023/12 – 2024/03	**Bluetooth Car Controlled** - Using ESP32 to controll 4 DC motor - 2 modes : Automation (avoiding osbtacles) and hand-controlled (adjusting car direction through bluetooth). - C Program (Arduino IDE) - The Mobile App controlls ESP32 through bluetooth - PID Control - ultrasonic sensor
2025/01 – 2024/05	**Using geophone in seimic detection** - Convert geophone wave to digital signal through ADC và analyze data by stm32. - op-amp amplify the geophone signal after filtering
2025/02 - 2025/05	**Countdown circuit using Atmega16** - Circuit use external interrupt, Timer1 Interrupt, 7-segment display and update the values of hours, minutes, seconds, tiktaks, EEPROM - AVR program (C/C++) - Coutdown and reset button - Saving the current value of 7-segment displays to EEPROM to save data when electric is off
2025/02 – 2025/05	**Room tempurature controll using ESP32**

	- Circuit use PID controll to adjust environment tempurature surrouding dht11 - Keep the tempurate in tempurature setpoint - Used an incandescent bulb as a heat source - controlled fan to adjust surrounding temperature using PID and BJT - Measured temperature with DHT11 sensor - included a keypad for inputting the desired temperature
2025/09 – 2026/01	**Light Control and Fan Control** - **Tanh Algorithm + I Controller to get Light Setpoint smoothly, same as fan control** - **Read DHT22 using 40-bit frame** - **LDR with voltage divider** - **Winforms App to visualize the value of tempurature, humidity, ADC LDR and set a setpoint** - **Firebase Database**